Dr. Torsten Busch
Tel.: +49 (0) 69/257203 012
Fax: +49 (0) 69/257203 999
Email: torsten.busch@curiaglobal.com

Curia Germany GmbH
Process Research & Development
Industriepark Hoechst, D569
D-65926 Frankfurt am Main

May 21st 2024

Graphene synthesis project update

1. Biochar synthesis

- Overall seven syntheses finished with hemp fibers from BAFA neu GmbH
  - three syntheses done with 0.02M sulfuric acid
    - first product not representative, not used further
    - second and third batch converted (see below)
  - four syntheses done with 0.10M sulfuric acid
    - very similar observations and outputs
    - products of these have been combined/homogenized
    - part of the combined lots has been converted

- Canadian Rockies Hemp arrived and cleared customs
  - first five synthesis batches finished (7.8 g product in total), one more in progress

- (Re)-Activation of larger vessel in progress

2. Biochar analysis

- No samples submitted yet

3. Graphene synthesis

- two experiments finished with steep heating ramp (20-30 °C/min)
  - biochar ground manually for first experiment
  - laboratory mill used for second experiment
  - products not yet characterized
  - no serious handling issues
- two experiments performed with much gentler heating ramp (3 °C/min)
  - better handling but no other apparent difference

4. Graphene analysis

- First samples (TB1133, Sigma-Aldrich Reference) sent out for Raman / SEM
  - Results expected this week
- Very promising discussion with one potential service provider (TEM, BET) today, expecting offer and NDA draft

5. Currently open questions

- Milling of material necessary/beneficial before TEM measurement?
- What type of TEM grid should be used?

Experiment overview

Experiment	Reactor	Raw material	Acid	T	t	pressure	Wash	Output		Experiment	Qty	Base	grinding	Gas	T (rate)	T (max)	t	Wash	Drying	Output
MB2928	AV1	6.0 g BAFA neu Hemp Fibre VF	100.0 g 0.19% 0.02 M Sulfuric acid	180 °C	22.5 h	8.7 bar -> 10.4 bar	742 g Water	2.6 g	->											
Comment	Fibres not completely covered by liquid																			
MB2933	AV1	6.1 g BAFA neu Hemp Fibre VF	174.0 g 0.19% 0.02 M Sulfuric acid	180 °C	24 h	8.7 bar -> 9.5 bar	647 g Water	1.2 g	->	TB1131	1.2 g	1.2 g KOH (90%)	manual	Ar	20-27 °C/min	790 °C	1 h	20 g 10% HCl	100 °C N2 stream	0.16 g
Comment	dark crusts on glassware, metal and liquid surface removed before filtration									Comment	ground biochar: brown powder, strong electrostatic charge							(+ water)	(atm. Pressure)	
MB2935	AV1	6.0 g BAFA neu Hemp Fibre VF	172.9 g 0.19% 0.02 M Sulfuric acid	180 °C	23 h	8.8 bar -> 9.5 bar	500 g Water	1.3 g	->	TB1132	1.3 g	1.3 g KOH (90%)	mill (2 min)	Ar	20-29 °C/min	792 °C	1 h	20 g 10% HCl	100 °C N2 stream	0.3 g
Comment	dark crusts on glassware, metal and liquid surface removed before filtration									Comment	ground biochar: brown powder							(+ water)	(atm. Pressure)	
MB2936	AV1	6.0 g BAFA neu Hemp Fibre VF	172.4 g 0.96% 0.10 M Sulfuric acid	180 °C	23 h	8.8 bar -> 10.6 bar	589 g Water	0.8 g		TB1133	1.0 g	1.0 g KOH (90%)	mill (1 min)	Ar	3 °C/min	785 °C	1 h	20 g 10% HCl	100 °C N2 stream	0.3 g
Comment	dark crusts on glassware, metal and liquid surface removed before filtration									Comment	ground biochar: brown powder; see temperature trend							(+ water)	(atm. Pressure)	
MRa225	AV1	6.0 g BAFA neu Hemp Fibre VF	153.8 g 0.96% 0.10 M Sulfuric acid	180 °C	24 h	? -> 10.5 bar	500 g Water	0.9 g		TB1134	1.0 g	1.0 g KOH (90%)	mill (2 min)	Ar	3 °C/min	771 °C	1 h	20 g 10% HCl	100 °C N2 stream	0.3 g
Comment	dark crusts on glassware, metal and liquid surface removed before filtration								->	Comment	ground biochar: brown powder; see temperature trend							(+ water)	(atm. Pressure)	
MRa228	AV1	6.3 g BAFA neu Hemp Fibre VF	154.0 g 0.96% 0.10 M Sulfuric acid	180 °C	22.5 h	8.7 bar -> 11.0 bar	500 g Water	0.9 g												
Comment	dark crusts on glassware, metal and liquid surface removed before filtration																			
MRa231	AV1	6.1 g BAFA neu Hemp Fibre VF	168.8 g 0.96% 0.10 M Sulfuric acid	180 °C	23.5 h	8.0 bar -> 10.5 bar	500 g Water	0.8 g												
Comment	dark crusts on glassware, metal and liquid surface removed before filtration																			
MB2937	AV1	9.4 g Canadian Rockies Hemp	179.3 g 0.96% 0.10 M Sulfuric acid	180 °C	24 h	8.6 bar -> 10.9 bar	569 g Water	1.6 g	->											
Comment	very low amount of foreign material observed and removed during filtration																			
MB2938	AV1	8.9 g Canadian Rockies Hemp	175.0 g 0.96% 0.10 M Sulfuric acid	180 °C	23.5 h	9.2 bar -> 13.2 bar	577 g Water	1.4 g	->											
Comment	very low amount of foreign material observed and removed during filtration																			
MB2940	AV1	9.2 g Canadian Rockies Hemp	177.1 g 0.96% 0.10 M Sulfuric acid	180 °C	22.5 h	8.7 bar -> 10.6 bar	555 g Water	1.5 g	->											
Comment	very low amount of foreign material observed and removed during filtration																			
MB2942	AV1	8.9 g Canadian Rockies Hemp	177.6 g 0.96% 0.10 M Sulfuric acid	180 °C	23 h	8.3 bar -> 13.5 bar	578 g Water	1.6 g	->											
Comment	low amount of foreign material observed and removed during filtration																			
MB2943	AV1	9.8 g Canadian Rockies Hemp	178.2 g 0.96% 0.10 M Sulfuric acid	180 °C	24 h	8.3 bar -> 12.7 bar	611 g Water	1.7 g	->											
Comment	low amount of foreign material observed and removed during filtration																			